Study guide for Exam PL-300: Microsoft Power BI Data Analyst

Tip

- A new Practice Assessment experience is available on AI Skills Navigator that more closely aligns with the exam experience.
- To try it, SIGN IN to AI Skills Navigator at Microsoft Certified: Power BI Data Analyst Associate.
- Launch the Practice Assessment.
- You can also take the legacy Practice Assessment on Microsoft Learn at https://aka.ms/PL300-LearnPA. Please note that legacy Practice Assessments on Learn are no longer being updated. We strongly recommend using the new experience.

Purpose of this document

This study guide should help you understand what to expect on the exam and includes a summary of the topics the exam might cover and links to additional resources. The information and materials in this document should help you focus your studies as you prepare for the exam.

Expand table

Useful links	Description
Certification renewal	Microsoft associate, expert, and specialty certifications expire annually. You can renew by passing a **free** online assessment on Microsoft Learn.
Your Microsoft Learn profile	Connecting your certification profile to Microsoft Learn allows you to schedule and renew exams and share and print certificates.
Exam scoring and score reports	A score of 700 or greater is required to pass.
Exam sandbox	You can explore the exam environment by visiting our exam sandbox.

Useful links	Description
Request accommodations	If you use assistive devices, require extra time, or need modification to any part of the exam experience, you can request an accommodation.

Updates to the exam

Our exams are updated periodically to reflect skills that are required to perform a role. We have included two versions of the Skills Measured objectives depending on when you are taking the exam.

We always update the English language version of the exam first. Some exams are localized into other languages, and those are updated approximately eight weeks after the English version is updated. Although Microsoft makes every effort to update localized versions as noted, there may be times when the localized versions of an exam are not updated on this schedule. Other available languages are listed in the **Schedule Exam** section of the **Exam Details** webpage. If the exam isn't available in your preferred language, you can request an additional 30 minutes to complete the exam.

Note

The bullets that follow each of the skills measured are intended to illustrate how we are assessing that skill. Related topics may be covered in the exam.

Note

Most questions cover features that are general availability (GA). The exam may contain questions on Preview features if those features are commonly used.

Skills measured as of April 20, 2026

Audience profile

As a candidate for this exam, you should deliver actionable insights by working with available data and applying domain expertise. You should:

- Provide meaningful business value through easy-to-comprehend data visualizations.

- Enable others to perform self-service analytics.

As a Power BI data analyst, you work closely with business stakeholders to identify business requirements. You collaborate with analytics engineers and data engineers to identify and acquire data. You use Power BI to:

- Prepare the data
- Model the data
- Visualize and analyze data
- Manage and secure Power BI

You should be proficient at using Power Query and Data Analysis Expressions (DAX).

Skills at a glance

- Prepare the data (25–30%)
- Model the data (25–30%)
- Visualize and analyze the data (25–30%)
- Manage and secure Power BI (15–20%)

Prepare the data (25–30%)

Get or connect to data

- Identify and connect to data sources or a shared semantic model
- Change data source settings, including credentials and privacy levels
- Choose between DirectLake, DirectQuery, and Import
- Create and modify parameters

Profile and clean the data

- Evaluate data, including data statistics and column properties

- Resolve inconsistencies, unexpected or null values, and data quality issues
- Resolve data import errors

Transform and load the data

- Select appropriate column data types
- Create and transform columns
- Group and aggregate rows
- Pivot, unpivot, and transpose data
- Convert semi-structured data to a table
- Create fact tables and dimension tables
- Identify when to use reference or duplicate queries and the resulting impact
- Merge and append queries
- Identify and create appropriate keys for relationships
- Configure data loading for queries

Model the data (25–30%)

Design and implement a data model

- Configure table and column properties
- Implement role-playing dimensions
- Define a relationship's cardinality and cross-filter direction
- Create a common date table
- Identify use cases for calculated columns and calculated tables

Create model calculations by using DAX

- Create single aggregation measures

- Use the CALCULATE function
- Implement time intelligence measures
- Use basic statistical functions
- Create semi-additive measures
- Create a measure by using quick measures
- Create calculated tables or columns
- Create calculation groups

Optimize model performance

- Improve performance by identifying and removing unnecessary rows and columns
- Identify poorly performing measures, relationships, and visuals by using Performance Analyzer and DAX query view
- Improve performance by reducing granularity

Visualize and analyze the data (25–30%)

Create reports

- Select an appropriate visual
- Format and configure visuals
- Create a narrative visual with Copilot
- Apply and customize a theme
- Apply conditional formatting
- Apply slicing and filtering
- Use Copilot to create a new report page
- Use Copilot to suggest content for a new report page

- Configure the report page
- Choose when to use a paginated report
- Create visual calculations by using DAX

Enhance reports for usability and storytelling

- Configure bookmarks
- Create custom tooltips
- Edit and configure interactions between visuals
- Configure navigation for a report
- Apply sorting to visuals
- Configure sync slicers
- Group and layer visuals by using the Selection pane
- Configure drillthrough navigation, including pages, filters, and buttons
- Configure export settings
- Design reports for mobile devices
- Enable personalization in a report, including personalized visuals
- Design and configure Power BI reports for accessibility
- Configure automatic page refresh

Identify patterns and trends

- Use the Analyze feature in Power BI
- Use grouping, binning, and clustering
- Use AI visuals
- Use reference lines, error bars, and forecasting

- Detect outliers and anomalies
- Use Copilot to summarize the underlying semantic model

Manage and secure Power BI (15–20%)

Create and manage workspaces and assets

- Create and configure a workspace
- Configure and update an app
- Publish, import, or update items in a workspace
- Create dashboards
- Choose a distribution method
- Configure subscriptions and data alerts
- Promote or certify Power BI content
- Identify when a gateway is required
- Configure a semantic model scheduled refresh

Secure and govern Power BI items

- Assign workspace roles
- Configure item-level access
- Configure access to semantic models
- Implement row-level security roles
- Configure row-level security group membership
- Apply sensitivity labels

Study resources

We recommend that you train and get hands-on experience before you take the exam. We offer self-study options and classroom training as well as links to documentation, community sites, and videos.

Expand table

Study resources	Links to learning and documentation
Get trained	Choose from self-paced learning paths and modules or take an instructor-led course
Find documentation	Power BI documentation Microsoft Power Apps documentation
Ask a question	Microsoft Q&A | Microsoft Docs
Get community support	Power Apps - Power Platform Community Power Query - Power Platform Community Building Power Apps - Power Platform Community
Follow Microsoft Learn	Microsoft Learn - Microsoft Tech Community
Find a video	Watch the Exam Prep video #LessCodeMorePower | Shows Browse other Microsoft Learn shows

Change log

The table below summarizes the changes between the current and previous version of the skills measured. The functional groups are in bold typeface followed by the objectives within each group. The table is a comparison between the previous and current version of the exam skills measured and the third column describes the extent of the changes.

Expand table

Skill area prior to April 20, 2026	Skill area as of April 20, 2026	Change
Audience profile		No change
Prepare the data	**Prepare the data**	No change
Get or connect to data	Get or connect to data	Minor

Skill area prior to April 20, 2026	Skill area as of April 20, 2026	Change
Visualize and analyze the data	**Visualize and analyze the data**	No change
Enhance reports for usability and storytelling	Enhance reports for usability and storytelling	Minor
Manage and secure Power BI	**Manage and secure Power BI**	No change
Create and manage workspaces and assets	Create and manage workspaces and assets	Minor

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